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History of Astronomy: Einstein's Revolution

Relativity Changes Everything

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Introduction

In 1905 a clerk in a Swiss patent office published a paper that quietly removed two of the pillars of physics: absolute space and absolute time. In 1915, after a decade of the hardest intellectual labor in the history of the subject, the same man published a theory that replaced gravity itself with the geometry of the universe. This book tells the story of that double revolution, Einstein's revolution, not as biography but as astronomy: how the theory was forced into being by the observations and the failures of the nineteenth century, how it was first tested with the light of eclipses and the orbit of an errant planet, and how the century that followed turned relativity from a controversial abstraction into the working language of the science of the sky.

The revolution's shape can be held in one sentence. Before Einstein, physics described a universe of things moving through space and time, which were themselves fixed and indifferent; after him, physics describes a universe of fields and events woven into a single fabric, spacetime, whose shape is gravity and whose behavior every large-scale phenomenon in astronomy now expresses. The change was not a small correction to the old physics but a change of foundation, and its consequences reached every corner of the science: the orbits of planets, the light of stars, the fate of collapsing suns, the expansion of the universe, and the waves that ripple through space itself, all became problems written, by default, in relativistic terms.

What follows is a book about the revolution rather than the revolutionary. Einstein appears throughout, as the theory's author and its most persistent critic, but the chapters keep their eye on the idea and its life in astronomy. The narrative runs from the cracks in Newton's universe, through the two postulates of 1905 and the geometry of 1915, to the eclipse that made the theory famous and the century of tests that made it unavoidable, and closes with the instruments of the present era, which detect the theory's predictions as a matter of daily routine. It is the story of how astronomy absorbed the most abstract physics ever written and became, in the process, the science that measures the shape of space and time themselves.

A note on what the reader will not find may prevent a wrong expectation. This is not a biography: Einstein's life appears only where the theory's story requires it, the patent office, the wartime eclipse, the later decades of debate, and the personal history, the marriages, the politics, the music, belongs to other books in the library. Nor is it a textbook: the mathematics is described in words, the equations appear only as ideas, and the reader is asked for patience rather than background. What the book offers instead is the connected narrative the specialist treatments usually omit, the path from one astronomer's anomaly to another astronomer's photograph, and the reader who finishes it should hold the theory's whole arc in mind, from its founding paradoxes to its routine daily use in the sky's own instruments, which is what a revolution looks like once it has won. The chapters that follow assume no prior physics, quote no equations, keep the mathematics to its ideas, and promise the reader one thing the specialist treatments rarely offer: the whole story, from the first crack in the old confidence to the newest instrument now listening for the geometry's next word, told as the connected history it actually was.

A note on the word revolution, because the volume's title promises a specific claim and the account should define it. A scientific revolution, in the sense used here, is a change in the background assumptions that ordinary work runs on: the fixed stage of space and time was such an assumption, used nightly by astronomers and daily by engineers, and its replacement by a stage that bends and stretches is the event this book narrates; the account measures the claim by consequences rather than rhetoric throughout, counting the instruments, the observations, and the engineering that only makes sense if the revolution is real, and the reader who distrusts the word may substitute the plainest available sentence: the old background failed a series of tests, and the new background has passed a century of them.

A note on the mathematics the book refuses, because the reader deserves the account's own boundaries stated early. The general theory is written in a compact notation that takes years to read fluently, and the chapters do not pretend to teach it; instead they use the theory's geometry, the rubbery stage, the slow clock, the bent ray, the straightest path through curved spacetime, images the physics itself licenses and the experiments confirm; where a number matters the account gives the number, and where a picture suffices the account gives the picture, the reader being promised nothing that requires a blackboard and given everything that requires attention.

Chapter 1: The Discovery of Einstein's Revolution

Every revolution has its eve, and the eve of this one was the most confident period in the history of physics. By the close of the nineteenth century, Newton's mechanics and Maxwell's theory of light seemed to have divided the

physical world between them, and more than one prominent physicist advised students that the remaining work was measurement in the fourth decimal place. The confidence was honest, because the edifice worked: planets followed Newton's law to the precision of the finest instruments, and light followed Maxwell's equations through every experiment the laboratories could devise. The cracks, when they came, came not from carelessness but from precision, from experimenters who kept pushing into the fourth decimal place and found numbers that the edifice could not produce.

The first crack was the most famous null result in physics. If light was a wave in a medium, the luminiferous ether, then the Earth, moving through that medium on its orbit, should show different light speeds in different directions, as a swimmer's speed relative to shore differs across and along the current. The experiment designed to detect that difference, refined over years and executed with exquisite care in 1887, found nothing: light moved at the same speed in every direction, in every season, as if the medium that was supposed to carry it did not exist. The result was not a small anomaly but a contradiction, and the following two decades produced a series of patchwork responses, contraction hypotheses and modified ethers, that preserved the old mathematics at the cost of ever stranger assumptions, a program whose unease its own practitioners made no effort to hide.

The second crack was subtler and came from the equations themselves. Maxwell's theory gave light a definite speed, but no experiment or analysis could establish what that speed was measured against: the ether, unobservable; the source, as sound is measured against its emitter; or something deeper. Theoretical work near the century's turn showed that the equations of electromagnetism kept their form under a particular set of transformations, and that those transformations mixed space with time in a way no one had been willing to take literally. The mathematics was in place, in other words, before anyone would own it; what was missing was the willingness to say that lengths and durations themselves are not absolute but relative, and that the mixing was not a trick of the formalism but a fact about the world.

The revolution's discovery, then, was an act of interpretation. In 1905, Albert Einstein, then working in the Swiss patent office, took the two plainest statements the situation allowed, that the laws of physics take the same form for every observer moving at constant speed, and that light travels at the same speed for every such observer, and followed them without compromise. The results he drew, moving clocks run slow, moving lengths contract, simultaneity is a matter of viewpoint, and mass is a form of energy, were not refinements of the ether program but its replacement, and the physics community spent the following decade absorbing what it meant for a foundation of the subject to be rebuilt by a clerk who had asked, simply, what the experiments had already said. The chapters that follow trace that rebuilding and its consequences, through the generalization to gravity a decade later and the astronomical century that turned the theory's predictions into observations.

The discovery's most remarkable feature is what it left out. Einstein built the special theory without the machinery his predecessors considered essential, no ether to define the light's frame, no experimental program of his own, and no mathematics beyond the algebra and patience of a determined outsider, and the paper that changed physics was written, as its author later noted, in the spare hours of a desk job. The absence is not trivia but method: the ether had become the subject of ever more elaborate rescue attempts, and the young theorist's advantage was precisely his willingness to let the unobservable thing go. The lesson entered the practice of the century that followed, in which the unobservables of a theory are treated with suspicion and its invariants are honored, and the 1905 papers, four of them in one year, remain the standing example of how much a field can be moved by a single well-aimed question asked at the right moment of its crisis.

The eve of the revolution also owed a debt to astronomy itself, which the century's confidence tended to forget. The Newtonian edifice had been built from the sky: Kepler's laws were its founding observations, and every refinement of the theory had been checked against the finest observations of the age, the planets, the comets, the tides. But the same pursuit of precision that validated the theory also exposed its edges, and by the nineteenth century's end the most precise instruments of the age, the interferometers that could measure light's speed to exquisite exactness, were reporting a number the theory could not interpret, the same number in every direction. The revolution's lesson about instruments deserves stating plainly, because it recurs through this book: every decisive turn in the theory's history came from an instrument pushed past its design, a Michelson interferometer, an eclipse camera, a radio dish, a wave detector's mirror, and the reader should watch for the pattern, precision first, anomaly second, revolution last, whenever a chapter's story turns.

The office's irony wants its room in the discovery chapter, because the revolution's birthplace was a desk for examining other people's inventions. The Swiss patent office employed the young theorist as a technical examiner, the daily work being the reading of applications for electromagnetic devices, the synchronization of clocks by

signals being a live engineering question of the era; the chapter's honest speculation, flagged as such: the day job kept the physics of time and signals in front of the examiner for years, the trade's practical puzzles feeding the theorist's radical question, and the biography's lesson being that a revolution can be drafted on office time, the era's most confident physics unsettled by a man whose job was to check the correctness of patent claims.

The year of papers wants the second room, because the twelve months of 1905 are the revolution's first harvest and the account should count it. The same year that produced the special theory also produced the papers on the light quantum, the atomic reality of Brownian motion, and the equivalence of mass and energy, four works any one of which would carry a normal career; the discovery chapter keeps the count for its honest proportion: the revolution was not a single thunderbolt but a season of them, the clerk's annus producing a full dozen of the century's physics in ten months, and the reader who wants the revolution's texture finding it there: not one great paper but a young man thinking so fast the journals could barely keep up.